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CSC 101: Introduction to Computer Science - Hardware, Data and Algorithms

What actually happens inside a computer, and how to think in algorithms

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Computer Science | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

CSC 101 is often taught as a tour of parts, but the exam tests understanding. This pack explains the stored-program model, number systems, data representation, operating systems, networks and algorithmic thinking, with worked binary and flowchart exercises.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the stored-program architecture and the roles of CPU, memory and storage.
- Convert between binary, decimal and hexadecimal and perform simple binary arithmetic.
- Represent text, images and sound in digital form and explain the trade-offs.
- Design a simple algorithm as pseudocode and a flowchart, and trace it for correctness.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The stored-program model

Modern computers follow the idea that instructions and data live in the same memory and are fetched and executed in sequence.

- CPU: control unit fetches and decodes, arithmetic logic unit executes.
- Registers: small, fast storage inside the CPU.
- Main memory: volatile, holds running programs and data.
- Secondary storage: persistent, slower, larger.
- Input and output devices move data between the machine and the world.

- The fetch-decode-execute cycle repeats for each instruction.

Number systems and data representation

Computers store patterns of bits. Everything else is a convention layered on top.

- Binary to decimal: sum each bit times its place value.
- Decimal to binary: divide repeatedly by 2 and read remainders upward.
- Hexadecimal groups four bits and is used for compact memory addresses and colours.
- One byte is eight bits; ASCII uses seven or eight bits per character, Unicode uses variable encoding.
- Images are stored as pixel values; resolution and colour depth trade quality against size.
- Sound is sampled at intervals; sampling rate and bit depth determine fidelity and file size.

> **Practice conversion by hand**
> Exams rarely allow calculators for number-system questions. Convert 156 to binary and back until it takes under a minute.

Algorithms, pseudocode and flowcharts

An algorithm is a finite, unambiguous sequence of steps that solves a problem.

- Sequence, selection and iteration are the three control structures.
- Pseudocode should be readable by a human and precise enough to implement.
- Flowchart symbols: oval for start and end, parallelogram for input/output, rectangle for process, diamond for decision.
- Trace a loop by writing a table of variable values for each iteration.
- Test with normal, boundary and invalid inputs.
- An algorithm that produces an answer but never terminates is not correct.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Convert 45 to binary.

Answer: 45 = 32 + 8 + 4 + 1 = 101101 in binary.

2. Why is hexadecimal used in computing?

Answer: Because each hex digit maps exactly to four binary bits, making long binary values shorter and easier to read.

3. List the three basic control structures.

Answer: Sequence, selection (decision) and iteration (loops).

4. Write pseudocode to find the largest of three numbers.

Answer: Read a, b, c. Set max = a. If b > max then max = b. If c > max then max = c. Print max.

5. What happens during the fetch-decode-execute cycle?

Answer: The CPU fetches the next instruction from memory, decodes what operation it specifies, executes it, and stores any result before repeating.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Convert ten numbers between binary, decimal and hexadecimal without a calculator.
- [ ] Draw and label the components of a computer system.
- [ ] Write pseudocode for three simple problems and trace each with sample inputs.
- [ ] Draw one flowchart using correct symbols for a decision loop.
- [ ] Explain the difference between RAM and secondary storage in two sentences.
- [ ] Learn ten computing terms and define each without jargon.
- [ ] Practise one CSC 101 objective paper and review every error.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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